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RESEARCH ARTICLE

Enhancing Credit Risk Decision-Making in Supply Chain Finance With Interpretable Machine Learning Model

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ABSTRACT The increasing complexity of supply chain finance poses significant challenges to effective credit risk assessment. Traditional black-box models often fail to provide insights into the factors driving credit risk, which is essential for stakeholders when making informed decisions. By conducting analysis of interpretable machine learning models, the study evaluated their performance in assessing credit risks. Specifically, we applied Extreme Gradient Boosting (XGBoost), Random Forest (RF), Least Squares Support Vector Machine (LSSVM) and Convolutional Neural Network (CNN) models for risk assessment. Our methodology included an ablation experiment along with utilizing Shapley Additive Explanation (SHAP) to elucidate the contribution and significance of specific risk factors. The results indicated that the asset-liability ratio, cash ratio, and quick ratio notably influence credit risk. This study clarified the applicability and limitations of various models, highlighting the superior performance and interpretability of XGBoost through the SHAP algorithm. Ultimately, the insights from this study provided valuable guidance for companies and financial institutions, fostering more sustainable allocation of financial resources.

INDEX TERMS Supply chain finance, sustainable, credit risk, XGBoost, SHAP, machine learning.

I. INTRODUCTION

The global understanding of corporate credit risk is deeply influenced by a myriad of factors, each playing a crucial role in determining the financial stability of organizations. In recent years, China's financial landscape has experienced rapid developments, drawing increased attention to the broader economic structures and advancements within financial markets. A significant milestone was the inclusion of A-shares in the MSCI Emerging Markets Index in 2018, which has spurred considerable growth in risk management practices within the Chinese market. This evolution is not just specific to China but is reflective of broader trends where intricate and uncertain dynamics of supply chain finance (SCF) necessitate detailed scrutiny and innovative approaches [1]. Since corporate credit risk can lead to substantial losses for investors, effective credit risk management is crucial for maintaining economic stability. Accurately identifying corporate credit risk and enhancing the financing capacity of supply chain node enterprises has become increasingly important.

Disruptions in any supply chain node trigger ripple effects. Financial and credit risks may spread along the supply chain [2]. This study focuses on SCF risk assessment, utilizing advanced machine learning (ML) tools, including XGBoost, RF, LSSVM and CNN models. These models have various characteristics. XGBoost balances high performance and the ability to process large datasets. The capacity of RF to handle high-dimensional data and fend off overfitting is well recognized [3]. And it comprehensively considers the complex relationships between multiple factors. LSSVM performs well with limited samples [4]. It is suitable for certain situations where data acquisition is difficult. CNN is good at capturing spatial or sequential features in data [5].

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It is particularly effective in identifying potential risks in the supply chain. The SHAP method provides an explanation of the model's internal decision path, enhancing the accuracy and transparency of risk assessment.

The objective of this study is to generate insights that transcend traditional financial metrics, considering the long-term viability, resilience, and holistic stability of SCF participants. By exploring the strengths and limitations inherent in various models, the study aspires to amalgamate these strengths into a novel model that offers a more nuanced understanding of credit risks. Ultimately, this research aims to empower financial institutions with better tools for identifying and managing risks. Thereby fostering greater resilience within supply chains and enhancing their ability to withstand and adapt to unforeseen challenges.

The rest of this study is structured as follows: Literature review on SCF risk is presented in Section II. The model for evaluating credit risk is built in Section III. Section IV concentrates on the model performance comparison and interpretation. Section V provides implications, and Section VI presents conclusions.

II. LITERATURE REVIEW

A. IDENTIFICATION OF SCF RISK FACTORS

As an innovative financial service model, SCF is designed to solve the financing difficulties of small and medium-sized enterprises (SMEs). By targeting the entire supply chain as a financing object, it improves the liquidity of funds and the competitiveness of the supply chain. SMEs' credit risk is impacted by supply chain financing variables in addition to internal factors. SCF provides financing support to SMEs by integrating the financial flows, logistics, and information in the supply chain, but it also brings credit risks [6]. The credit risk assessment of SCF needs to comprehensively consider multiple dimensions such as industry conditions, operation conditions of SMEs and core enterprise, and supply chain relationship conditions [7]. Abbasi et al. [8] combined the distinctive qualities of IoT technology with the business procedure of inventory pledge financing model to design an IoT based SCF model. Trivedi [9] believed that the financial health status, production costs, sales strategies, and financial structure of a company can all affect its credit risk and decision-making in the supply chain. Changes in market interest rates, especially the rise in risk-free rates, can affect the present value of future cash flows and lower the worth of the company [10]. Xiao et al. [11] studied the application of blockchain technology in SCF by constructing an evaluation model, which included factors such as financing enterprises, core enterprises, asset financing status, blockchain platforms, and supply chain operations.

Maldonado et al. [12] proposed a cost-based feature selection SVM method, which considered feature acquisition costs through a mixed integer linear programming (MILP) model, thereby reduced feature acquisition costs while maintaining classification performance. Hajek and Michalak [13] compared different feature selection methods, including wrapper and filter methods, and found that wrapper method performed better in credit rating prediction. Arora and Kaur [14] proposed a consistent feature selection method based on Bootstrap Lasso, which can list and output selected features in one step through ensemble learning process and feature selection. Song and Peng [15] proposed a technique based on Multi Criteria Decision Making (MCDM) to evaluate imbalanced classifiers in credit and risk prediction by simultaneously considered multiple performance indicators.

B. RISK ASSESSMENT MODEL OF SCF

As big data and artificial intelligence technologies advance, methods for credit risk assessment and management are constantly innovating and improving. Xie et al. [16] adopted an improved AHP and Grey Aggregation method to determine the model weights. And they created a SCF credit evaluation system by thoroughly explaining the model's logical calculation issues and data flow. He et al. [17] proposed a group decision-making method based on hesitant information and applied it to corporate credit risk assessment. Huang et al. [18] demonstrated the operational and practical manner of the grey relational model in SCF risk assessment. Although these methods are flexible, they rely on subjective judgment and have limited accuracy [19], [20]. In order to improve the objectivity and accuracy of evaluations, modern financial credit risk models such as Credit risk+ and Portfolio view models are proposed, which utilized modern financial theory and placed greater emphasis on quantitative analysis [21]. However, there are still some issues with these models in practical applications, such as model assumptions that may be inconsistent with reality and the parameters are complex.

Including optimization models based on neural networks, SVM, and RF [22], advanced algorithms are used to handle complex nonlinear problems, improving the accuracy of credit risk assessment [23]. Liu et al. [24] analyzed the risks of international trade in SCF services in the energy industry using AI algorithms. Kim and Ahn [25] proposed a credit rating model based on multi-class support vector machine (MSVM), which improved classification performance through an ordinal pairwise partitioning (OPP) method and made computational resource utilization more efficient. Feng [26] analyzed the effectiveness of the SCF credit evaluation mechanism through intelligent algorithms and constructing intelligent evaluation models. Zhang et al. [7] used firefly algorithm to optimize SVM for credit risk evaluation in SCF. The result showed that FA-SVM is superior to LIBSVM in classification prediction accuracy. And it reduced the error rate of misjudging trustworthy enterprises as untrustworthy enterprises. Zhu et al. [6] proposed an enhanced hybrid ensemble machine learning method, which improved the accuracy of SME credit risk prediction by combining random subspace and Multi Boosting. This method demonstrated good performance in handling small sample data. The MLIA algorithm proposed by Ma and Lv [27] optimized the decision tree model by decomposing the objective function into weighted sums of base functions and using gradient boosting methods. Shen et al. [28] combined an improved synthetic minority oversampling technique (SMOTE) and LSTM network, as well as an AdaBoost algorithm for credit risk assessment. Deep learning models are usually more competitive than other models when dealing with imbalanced credit risk assessment problems.

C. AN APPRAISAL ON THE LITERATURE

Supply chain finance significantly improves capital mobility and competitiveness of supply chain by integrating capital flows, logistics and information flows. However, this process is also accompanied by credit risk, and its assessment needs to consider multiple dimensions such as industry, SMEs, core enterprises and supply chain relationships. With the application of IoT, blockchain and AI, new methods for SCF credit risk assessment have been provided. Optimization models based on neural networks, SVM and RF significantly improve the accuracy of evaluation by dealing with complex nonlinear problems. Previous studies focused on personal credit or single financial risk. The research gaps mentioned above need attention, this study integrates new technologies and focuses on driving factors, evaluates the credit risk of SMEs in the context of SCF. Thus, the enterprises and regulators can more effectively identify and manage risks in SCF and promote the health of the entire industry.

III. METHODOLOGY

A. DETERMINATION OF EVALUATION INDICATORS

The credit risk assessment of SMEs can be measured and estimated through multiple dimensions of evaluation criteria, such as the financial status of core enterprises and SMEs, supply chain indicators, and other non-financial related indicators. Through a detailed analysis of the literature, this study has obtained some preliminary indicators. Various operational activities of enterprises cannot be separated from financial support [29]. In terms of profitability, it is measured by operating profit margin and return on investment. Operational capacity is measured using total asset turnover, inventory turnover, etc. For solvency, the main measures are corporate financial indicators, such as current ratio and quick ratio.

In the context of SCF, supply chain related indicators serve as a bridge, connecting the relationship between SMEs and core enterprises. Therefore, this study evaluates credit risks through six indicators: industry sustainability, supply chain concentration, market prosperity index, regional GDP, degree of informatization, and ESG rating [30], [31]. **TABLE 1** shows the initial indicator system for credit risk assessment under SCF, with 25 selected initial indicators.

B. LEAST SQUARES SUPPORT VECTOR MACHINE (LSSVM)

SVM has been widely applied in various fields such as finance [32], healthcare [33], and environment [34] due to its global optimization capability and good robustness. Linear SVM exhibits faster training speed and lower computational complexity when processing large-scale datasets [35]. Yu et al. [36] used SVM in their studies and compared it with other classification techniques, demonstrating the effectiveness of SVM in credit risk assessment.

By introducing equality constraints and least squares linear equations, LSSVM transforms the quadratic programming problem in SVM into solving a system of linear equations. This transformation simplifies the calculation process, and improves the convergence speed and generalization ability of the algorithm [37].

There is a training set with N samples, $X=[X_1, X_2, \ldots, X_N]$ is the input data of the training samples, $Y=[Y_1, Y_2, \ldots, Y_N]$ is the output data of the training samples, the problem description of LSSVM is as follows:

$$min_{\omega,b,e}\, J(\omega, e) = \frac{1}{2}\omega^T\omega + \frac{\gamma}{2}\sum_{k=1}^{N} e_k^2 \tag{1}$$

$$s.t.\ y_k\left[\omega^T\varphi(x_k) + b\right] = 1 - e_k, \quad k = 1, 2, \ldots, N \tag{2}$$

where J is the optimization objective function; ω is the weight vector; e_k is the error term; γ is the regularization parameter; x_k is the input data for the k th training sample; y_k is the output data of the k th training sample; b is a random bias; $\varphi(x_k)$ is the kernel space mapping function.

Using Lagrange multiplier method to solve it [4].

$$L(\omega, b, e, \alpha) = J(\omega, e) - \sum_{k=1}^{N}\alpha_k\left\{y_k\left[\omega^T\varphi(x_k)+b\right]-1+e_k\right\} \tag{3}$$

where L is the Lagrangian function; α_k is the Lagrangian factor.

By taking partial derivatives of variables ω, b, e_k, and α_k in equation (3) and making the derivatives equal to zero, a set of linear equations is obtained. By solving this linear equation set, the LSSVM classification expression is obtained, as shown in equation (4):

$$y(x) = sgn[\sum_{k=1}^{N}\alpha_k y_k K(x, x_k) + b] \tag{4}$$

C. CONVOLUTIONAL NEURAL NETWORKS (CNN)

The CNN model was initially applied to image recognition, and in recent years, many scholars have applied convolutional neural networks to text recognition or data classification [38]. Moreover, CNN models can automatically learn features from the obtained data to reduce parameters and computational complexity. The powerful network structure of CNN is also more conducive to learning [39]. Following is an introduction to the structure of CNN.

1) CONVOLUTIONAL LAYER

Convolutional layers extract features from input data through convolution operations [51]. To connect the input layers

TABLE 1. Evaluation criteria of risks.

First level	Second level	Third level	Source
Qualification of Financing Enterprises	Profitability	Operating profit margin	Yin et al. [40]
		Return on equity	Luo et al. [41]
		Profitability of main business	Sang [42]
	Operational capability	Inventory turnover rate	Xiao et al. [11]
		Total asset turnover ratio	Gu et al. [19];Zhu et al. [6]
		Revenue growth rate	Yao et al. [43]
		Current ratio	Zhu et al. [6]
	Development capability	Total asset growth rate	Abbasi et al. [8];Yin et al. [40]
		Sales revenue growth rate	Abbasi et al. [8]
		Accounts receivable turnover rate	Zhang et al. [7]
	Debt paying ability	Assets and liabilities	Li et al. [44]
		Cash ratio	Li et al. [44]
		Quick ratio	Bi and Liang [45]
Core enterprise qualifications	Profitability	Return on equity	He et al. [17];Zhang et al. [7]
		Operating profit margin	Luo et al. [41]
		Quick ratio	Chen et al. [46];Gu et al. [19]
	Debt paying ability	Current ratio	Chen et al. [46];Gu et al. [19]
		Asset liability ratio	Chen et al. [46];Gu et al. [19]
		Profitability of main business	Yao et al. [43]
Supply chain operation situation	Macro-environment	Regional economic growth	Chen et al. [46]
		Market prosperity index	Xiao et al. [11]
		Industry sustainability rate	He et al. [17]
	Operation situation	Supply chain concentration	Yao et al. [43]
		Degree of informatization	Marcelin et al. [47];Xie [48]
		ESG rating	Deng et al. [49];Fu et al. [50]

through convolutional kernels, it is necessary to specify the size of the convolutional kernels.

2) POOLING LAYER

The pooling layer is used to reduce the dimensionality and complexity of feature maps, and preserving important feature information. The commonly used pooling methods include Max Pooling and Average Pooling [52]. The SMEs and corresponding core enterprise data have a large number of details and high algorithm complexity. The maximum pooling layer is ultimately used for pooling in this study.

3) FULLY CONNECTED LAYER

The fully connected layer is the output layer of the CNN model. It integrates the features extracted by previous convolutional layer and pooling layer, performs linear transformation through weights and biases. Then it performs nonlinear transformation through activation functions. The diagram of CNN is shown in **FIGURE 1**.

D. EXTREME GRADIENT BOOSTING (XGBoost)

XGBoost is an efficient decision tree-based ensemble learning method [53]. It seeks to improve the predictive performance of models by optimizing the goal function. The principle of XGBoost is to predict the new input data by weighted average of the results of multiple decision trees. It is superior to many ML algorithms in various estimation problems, and has high accuracy and model robustness [54]. In each decision tree, it automatically calculates the importance and weight of features, and selects the best features and nodes for splitting. In order to control the complexity of the model and prevent overfitting, regular terms are introduced

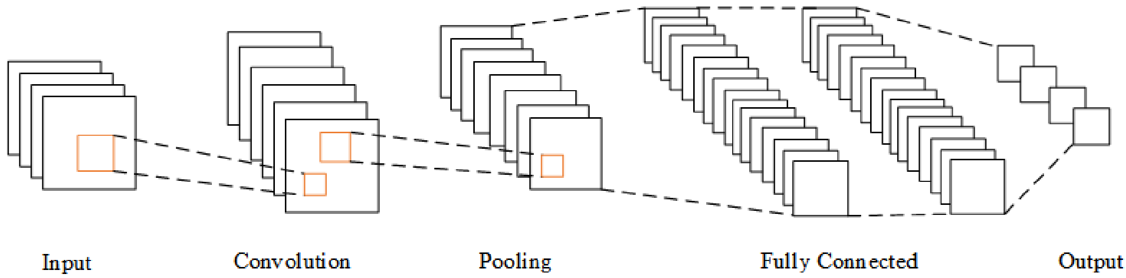

FIGURE 1. Structure of CNN.

into the objective function.

$$\Omega(f) = \gamma T + \frac{1}{2}\lambda \| \omega \|^2 \tag{5}$$

where γ and λ represent the penalty coefficient, and T is the number of leaf nodes.

XGBoost uses incremental training for iteration to maximize the reduction of loss function. After the t-round iteration, the objective function can be expressed as:

$$Obj^{(t)} = \sum_{i=1}^{n} l(y_i, \hat{y}_i^{(t-1)} + f_t(x_i)) + \Omega(f_t) \tag{6}$$

E. RANDOM FOREST (RF)

Random Forest Algorithm utilizes the idea of ensemble learning, integrates the classification results of multiple decision tree models to improve overall accuracy [55]. Feature selection is a crucial step aimed at identifying input feature. RF typically utilizes out of bag (OOB) methods to perform this process, helping optimize the model and reduce prediction errors, as shown in **FIGURE 2**. The basic idea of RF algorithm can be summed up as the subsequent steps:

(1) Use bootstrap sampling technique to randomly select several subsets of samples from the training set.

(2) Calculate classification nodes one by one from the selected features.

(3) Build multiple decision trees and select the optimal attributes for branching.

(4) The prediction results of the ensemble decision tree are used to determine the category of the new sample through voting.

F. SHAPLEY ADDITIVE EXPLANATION (SHAP)

SHAP is a method of explaining machine learning models based on additional feature attributes [56]. By integrating and computing the marginal contribution of feature subsets to the prediction result [57], the algorithm assigns a Shapley value to each feature. In this study, the model is interpreted by Tree SHAP, which uses a linear explanatory model to estimate the initial predictive model [58]. The key benefits of SHAP are its local and global interpretability, and the interaction value ensures the consistency of the interpretation of feature interaction effects. Compared to existing significance feature analysis in ML models, the SHAP method can determine if each input feature's contribution is favorable or negative. Its main idea comes from the Shapley value in combinatorial game theory. It is defined as the weighted average of the feature contribution of all possible feature combinators.

$$\varphi_i(f, x) = \sum_{S \subseteq N\{i\}} \frac{|S|!(M - |S| - 1)!}{M!} \times [f_x(S \cup f_i) - f_x(S)] \tag{7}$$

where φ_i is the Shapley value of the i th input value; N is the set of all features; M is the total number of input features; S is the subset of features used in the model; $f_x(S \cup f_i) - f_x(S)$ is the marginal contribution when subset S is added to feature i.

SHAP treats the output of the model as a linear sum of the input variables, and uses SHAP values to represent the contribution of each characteristic variable, the explanatory model $g(x')$ of the original model $f(x)$ can be expressed as:

$$f(x) = g(x') = \varphi_0 + \sum_{i=1}^{M} \varphi_i x_i' \tag{8}$$

IV. EMPIRICAL ANALYSIS

A. DATA PROCESSING

As a new financing solution, SCF is not widely used in China. Therefore, this study focuses on Chinese listed companies as a suitable dataset. All listed SMEs must have a trade relationship with one of the core enterprises in the supply chain. When SMEs have financing needs, they can seek financing from commercial banks with the participation of core enterprises. This study selected data from 851 listed companies as inputs for the model, with financial data from CSMAR and the Wind database. To solve the problem of data imbalance, the study initially fills missing values in the data sample. Following this, the SMOTE algorithm is employed for data preprocessing. **FIGURE 3** presents the diagram of SMOTE. The core concept of SMOTE lies in reclassifying minority samples to add the artificial simulated data, thereby mitigating the imbalance of dataset. Consequently, the accuracy and robustness of classification results are improved. **Table 2** shows the distribution of samples before and after oversampling.

TABLE 2. Data distribution before and after oversampling.

Category	Sample size			
	Before processing	Proportion	After processing	Proportion
Default	305	35.8%	546	50%
Normal	546	64.2%	546	50%
Total	851		1092	

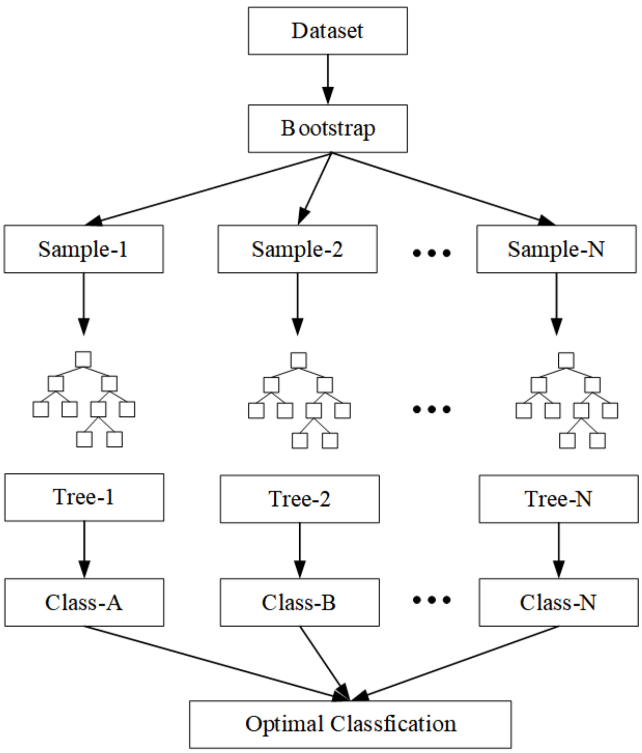

FIGURE 2. Flow chart of random forests.

B. DATA ANALYSIS

"Z-score" model considers multiple factors to measure the risk of corporate bankruptcy [59]. The Z-score is inversely proportional to the likelihood of a company experiencing financial crisis. For publicly traded manufacturing companies, when Z>2.99, the company's financial condition is good and the likelihood of bankruptcy is extremely low. According to the above criteria, the situation of whether the company is in default is classified.

FIGURE 4 depicts the correlation matrix of financial indicators in the data set. The current ratio, quick ratio and cash ratio are positively correlated with Z-score, while the current ratio has the highest positive correlation of 0.792. The negative correlation between Z-score and asset-liability ratio is high. And the absolute value is 0.819, which indicates that the increase of asset-liability ratio will lead to the increase of default risk of the company. In addition, several significant correlations were found between the independent variables. For example, the correlation between asset-liability ratio and the current ratio is −0.876, while the correlation between the quick ratio and the current ratio is 0.969. To reduce the amount of calculation and improve the interpretability of the model, the current ratio is considered to be deleted.

C. MODEL PERFORMANCE COMPARISON

In classification problems, there are two categories of samples: positive and negative. Companies with high default risk are labeled as positive categories, while companies with low default risk are labeled as negative categories. In addition to using accuracy to measure the proportion of correct predictions in the total sample, precision and recall can also be used as evaluation indicators [60].

Accuracy is defined as the proportion of all accurately predicted samples in the total sample:

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN} \tag{9}$$

Precision is defined as the proportion of default samples among the predicted default samples:

$$Precision = \frac{TP}{TP + FP} \tag{10}$$

Recall is defined as the proportion of correctly predicted default samples in the default sample:

$$Recall = \frac{TP}{TP + FN} \tag{11}$$

There is often a contradiction between precision and recall, in order to comprehensively consider these two indicators, F1-score is usually used as the evaluation metric, which is defined as the harmonic mean of precision and recall.

$$F1 - score = \frac{2 \times Precision \times Recall}{Precision + Recall} \tag{12}$$

As shown in the **FIGURE 5**, **FIGURE 6**, **FIGURE 7** and **FIGURE 8**, XGBoost model is superior to other models in all aspects in terms of model evaluation indicators, such as accuracy, precision, recall and F1-score. **TABLE 3** shows the descriptive statistical results for each indicator, the XGBoost model achieved an average precision of 92.8%, which is 2.6% better than the RF model, 7.1% better than the LSSVM model, and 7.5% better than the CNN model. The XGBoost model achieved an average accuracy of 91.9%, which is

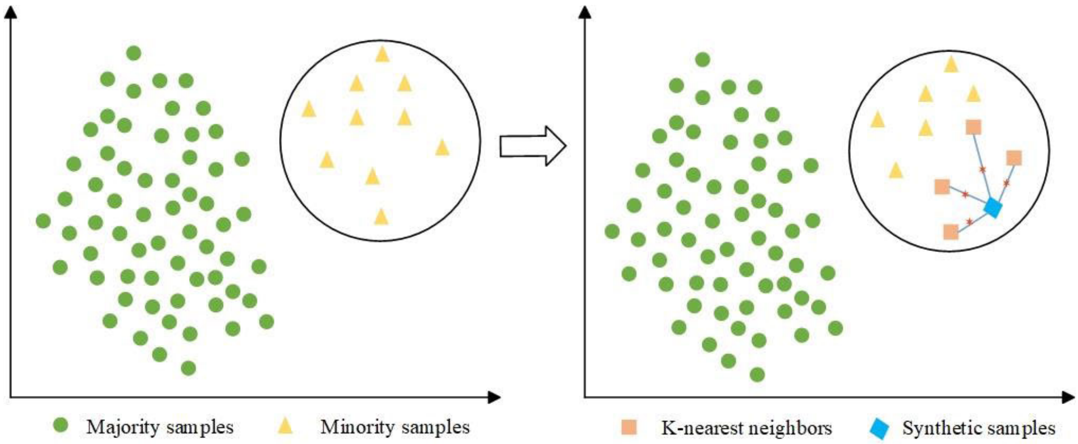

FIGURE 3. Synthetic samples generated from SMOTE.

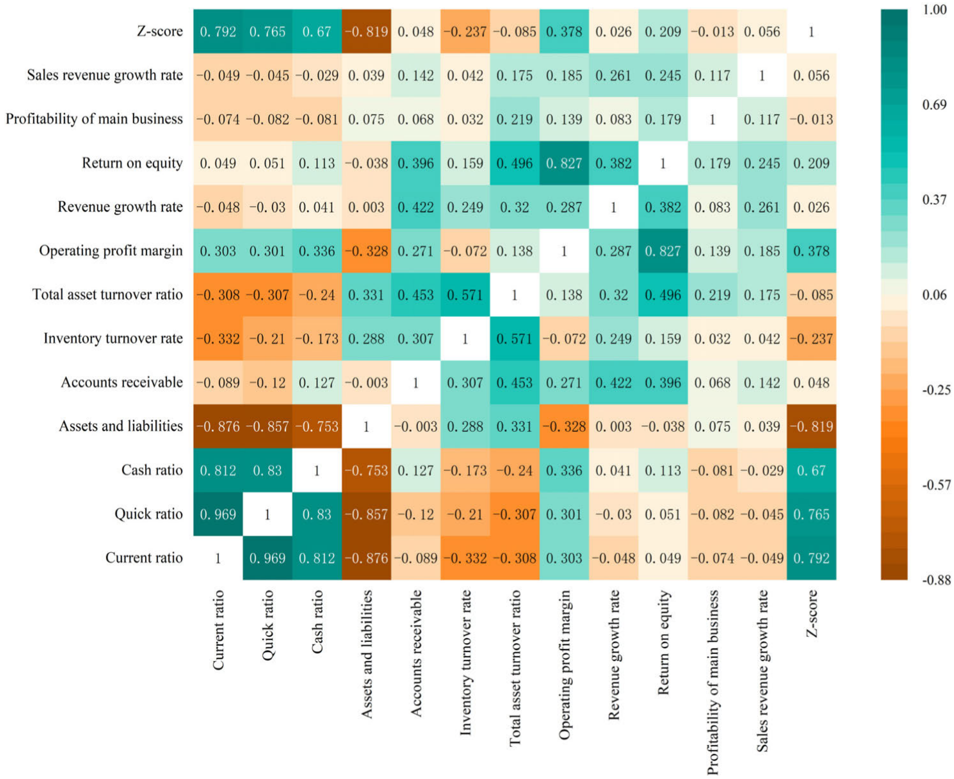

FIGURE 4. Heatmap of correlation coefficients.

4.2% better than the RF model, 7.5% better than the LSSVM model, and 9.3% better than the CNN model. In addition, the XGB's average recall rate reached 94.7%, which is 3.8%, 3.9%, and 7% higher than other models, respectively. Similarly, the XGB achieved an average F1-score of 93.7%, which is 3.2%, 5.6%, and 7.2% higher than the other models, respectively. It is concluded that XGBoost model demonstrates greater superiority in performance compared to the RF, CNN and LSSVM model.

The training and inference process of the CNN model requires numerous computational resources. And the low accuracy may be due to the high feature dimension and the small number of samples. The performance of LSSVM model depends on kernel function to a large extent. The selection of kernel function has a certain randomness, which makes it difficult to reach a high level of accuracy. While XGBoost can handle high-dimensional data without the need for feature selection, so it performs better in the risk assessment of this study. Based on the experimental conclusions, this study believes that the XGBoost model can more accurately evaluate the credit risk of SMEs.

D. SHAP INTERPRETATIONS

1) FEATURE CONTRIBUTION VALUE

For XGBoost model with better performance, SHAP method is used to compare the feature importance and contribution degree of the model results. The basic idea of SHAP values is to consider the average marginal contribution of features to the model's predicted value.

As shown in **FIGURE 9**, the horizontal axis is the SHAP value, and each point represents a sample. Blue denotes lower values and red denotes higher values. If a point is on the right side of the X-axis, it indicates that the value of the sample under this feature has a positive contribution to the output of the result, and it is more likely to be an enterprise with high credit risk.

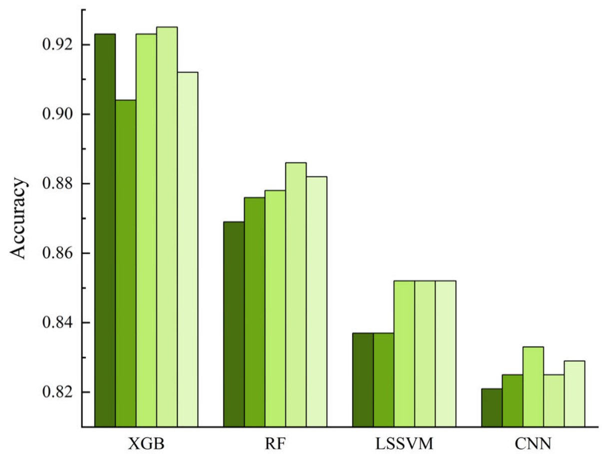

FIGURE 5. Accuracy comparison of models. Note: Each sample is represented by different color, same figure below.

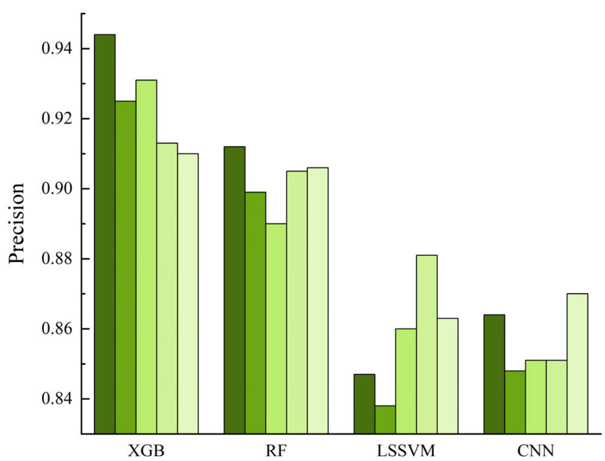

FIGURE 6. Precision comparison of models.

The asset-liability ratio is important for the prediction results. When the feature value is large, all the data points

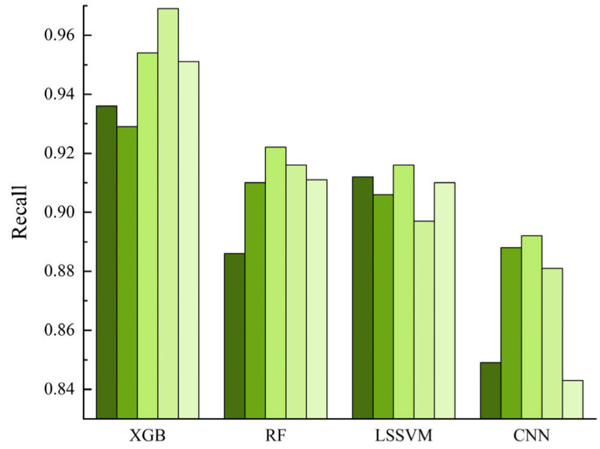

FIGURE 7. Recall comparison of models.

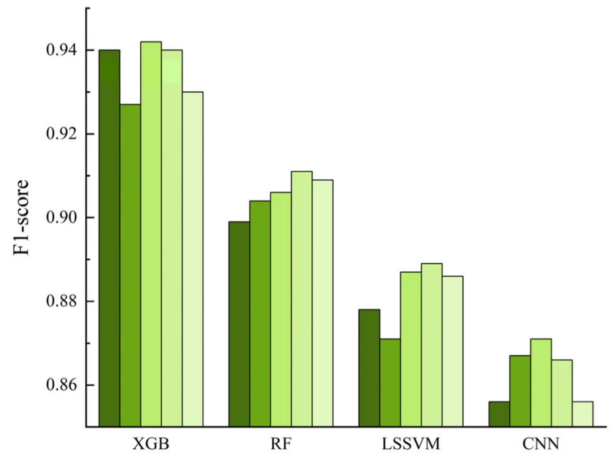

FIGURE 8. F1-score comparison of models.

are on the right side of the X-axis, indicating that when the asset-liability ratio is high, the chance of becoming defaulting enterprise increases.

In addition, the value of cash ratio and quick ratio has a clear influence on the result. The blue dots of these two indicators are mainly on the right, and the red dots are mainly on the left. It means the smaller the cash ratio and quick ratio, the more likely it is to be a default company. When the sample point is red, it means that the index value is large, but the corresponding SHAP value is less than 0, which indicate it has a negative effect on the model result, that is, the enterprise may be misjudged as a normal enterprise.

2) LOCAL INTERPRETATION

Force plot is an interpretation of a single sample. As shown in the **FIGURE 10**, the base value is the average of all sample estimates. Each feature variable influences the model's prediction through the SHAP value, thus pushing the baseline to the final prediction.

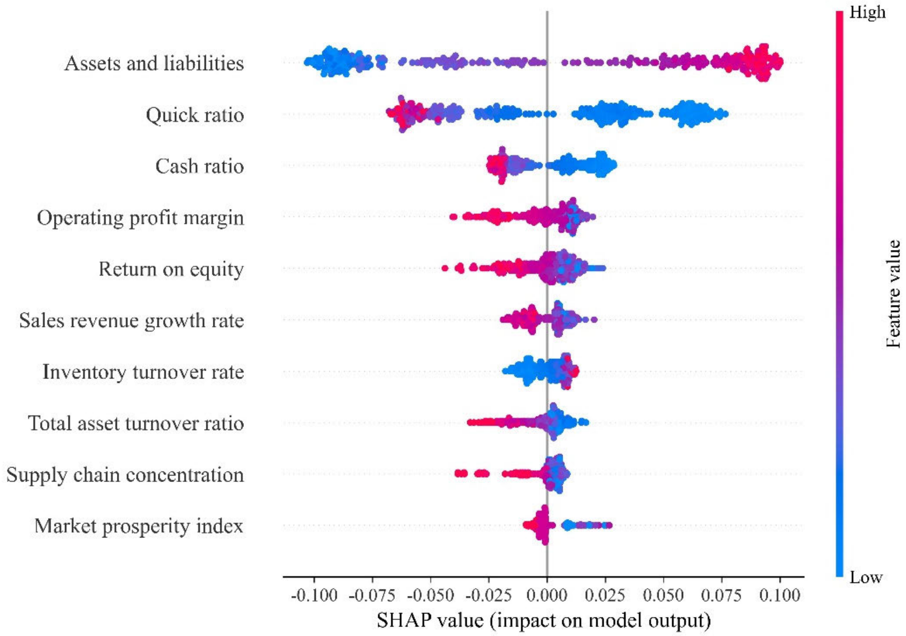

FIGURE 9. SHAP summary chart.

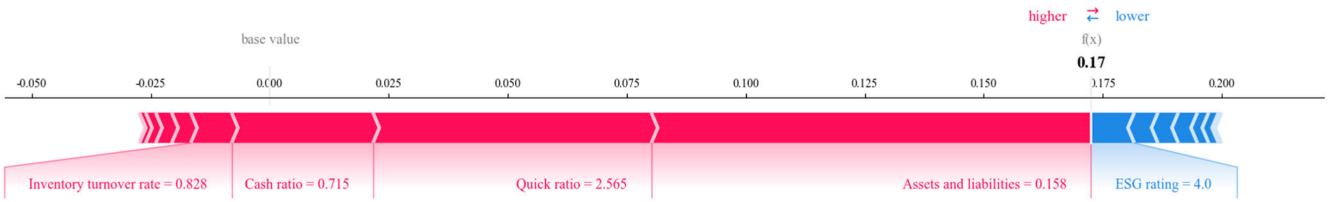

FIGURE 10. SHAP force plot of individual sample.

Red and blue indicate increases and decreases to the predicted result, respectively. The longer the arrow, the greater the influence of the feature on the output. Asset-liability ratio, cash ratio and quick ratio push up the prediction result. And they are the main driving factors for judging the sample as normal. The asset-liability ratio has the most significant influence on this sample. In contrast, the ESG rating pushes the predicted value down, prompting the model to classify the sample as a default company.

The contribution of the SHAP values of the remaining features to the model prediction is considered. It can be found that the combined effect of all characteristic variables finally raises the baseline of the prediction result, making the model classify the sample as normal one.

As depicted in **FIGURE 11**, the SHAP value escalates alongside the rising asset-liability ratio, indicating that the corporate credit risk increases. The asset-liability ratio reflects the enterprise's operational capacity fueled by creditor funds. Enterprise with high asset-liability ratio has a relatively weak debt repayment ability. Once there is a poor business situation or the market environment changes, it may fail to repay the principal and interest on time, disrupting capital chain and increasing credit risk.

In **FIGURE 12**, when the quick ratio exceeds 1.61, the corresponding SHAP value turns negative, signifying a decrease in default risk. Quick ratio is a vital index to measure an enterprise's ability to swiftly convert current assets into cash for immediate debt repayment. The quick ratio is a more accurate reflection of short-term solvency, because it excludes items such as inventory that takes longer to liquidate. Combined with the SHAP contribution value in **FIGURE 9**, the importance of asset-liability ratio and quick ratio ranks first and second respectively, emphasizing their significance in credit risk classification. Therefore, if the quick ratio is lower than 1.61, or the asset-liability ratio surpasses 0.36, the enterprise should pay attention to the potential credit risk.

Supply chain concentration significantly boosts enterprise risk taking. **FIGURE 13** show that when supply chain concentration exceeds 46.02, SHAP value becomes negative. ESG rating reflects the sustainable risks and opportunities

TABLE 3. Descriptive statistics of the indicator.

		XGB	RF	LS-SVM	CNN
Best	Accuracy	92.5	88.6	85.2	83.3
	Precision	94.4	91.2	88.1	87.0
	Recall	96.9	92.2	91.2	89.2
	F1-score	94.2	91.1	88.9	87.1
Mean	Accuracy	91.9	87.7	84.4	82.6
	Precision	92.8	90.2	85.7	85.3
	Recall	94.7	90.9	90.8	87.7
	F1-score	93.7	90.5	88.1	86.5
Variance	Accuracy	0.823	1.913	2.477	0.462
	Precision	0.411	0.683	1.902	0.203
	Recall	0.694	2.718	0.543	0.558
	F1-score	0.190	0.987	5.247	0.430

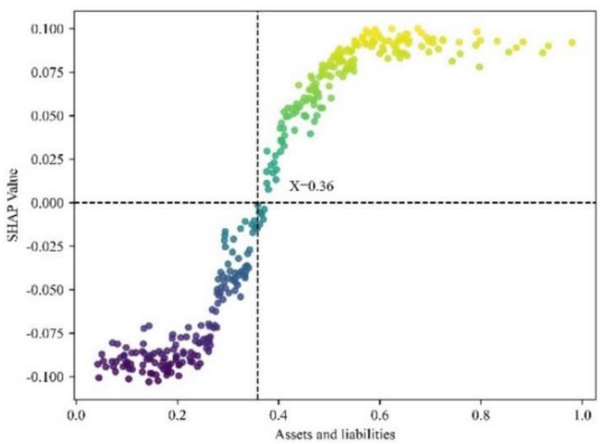

FIGURE 11. SHAP value for asset-liability ratio.

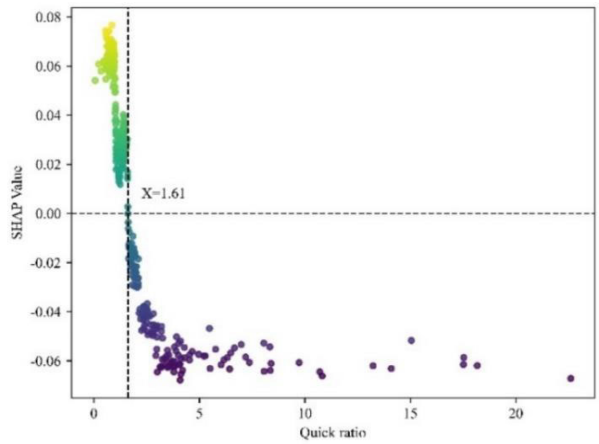

FIGURE 12. SHAP value for quick ratio.

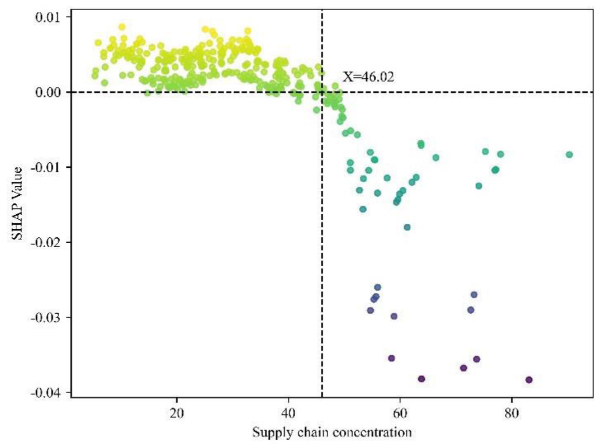

FIGURE 13. SHAP value for supply chain concentration.

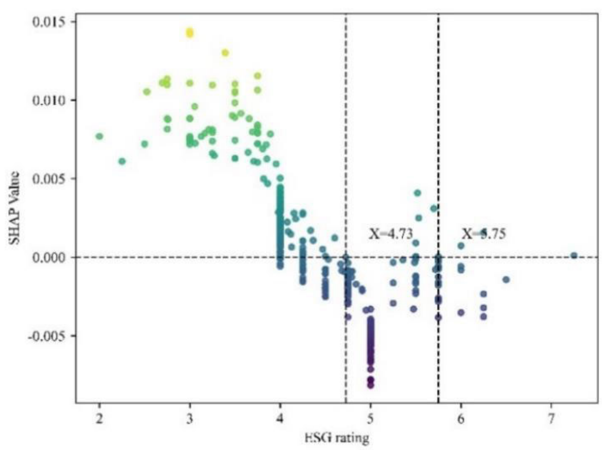

FIGURE 14. SHAP value for ESG rating.

of enterprises. In **FIGURE 14**, when the ESG rating is lower than 4.73, the enterprise risk level is increased. Therefore, when the supply chain concentration or ESG rating is reduced, it may represent the increase of credit risk.

3) ABLATION EXPERIMENTS

As shown in the **TABLE 4**, including the asset-liability ratio in the model improves performance compared to its exclusion. This underscores the pivotal role of the asset-liability ratio in credit risk evaluation, and it can provide crucial insights about the debt level and solvency of a company. The model's optimal performance is achieved when the quick ratio is integrated alongside the asset-liability ratio and cash ratio. This indicates that a concurrent consideration of these ratios enables a more comprehensively assessment of a company's financial position and credit risk.

Furthermore, SHAP analysis aligns with these findings by highlighting the significance of the asset-liability ratio, cash

TABLE 4. Ablation experiment results of the XGBoost model.

Features selection			Performance (%)		
Asset-liability ratio	Quick ratio	Cash ratio	Accuracy	Precision	F1-score
-	√	√	81.2 (±3.3)	85.1 (±5.2)	85.4 (±4.6)
√	-	√	85.8 (±1.5)	88.2 (±2.1)	89.0 (±1.9)
√	√	-	86.5 (±2.1)	88.3 (±3.5)	89.6 (±2.7)
√	√	√	91.5 (±1.1)	92.6 (±2.4)	93.4 (±0.7)

ratio, and quick ratio within the model. The performance changes observed in the ablation experiments mirror the outcomes of the SHAP analysis, confirming their substantial influence on the model's efficacy. Consequently, it is reasonable to conclude that the SHAP analysis results are accurate in this context, accurately pinpointing the features that are most crucial for credit risk evaluation.

V. IMPLICATION

Effective assessment of corporate credit risk in SCF has an extremely important impact on the healthy development of the entire supply chain. From the perspective of SMEs, they should establish a risk early warning mechanism, regularly monitoring changes in key indicators, such as cash ratio, quick ratio, asset-liability ratio. They should ensure sufficient short-term current assets, thereby maintaining good short-term solvency. At the same time, SMEs should focus on their sustainable development performance, including environmental protection, fulfilling social responsibilities, and improving corporate governance.

From the perspective of commercial banks, they should focus on cash ratio, quick ratio and asset-liability ratio of SMEs to comprehensively assess their solvency. Meanwhile, ESG rating should be incorporated into the risk assessment system as an important reference. They should reasonably determine the credit scale and term according to the actual situation to avoid risk exposure caused by excessive credit. Strengthen the post-loan management of SMEs, and regularly track and monitor changes in their business and financial status.

From the perspective of regulatory authorities, they should formulate and improve SCF supervision policies, clarify the role of financial indicators and performance ratings in risk assessment, and guide commercial banks and SMEs to pay attention to sustainable development. Enterprises are required to strengthen the disclosure of financial and performance information, improve the transparency and comparability of information.

VI. CONCLUSION

By combining machine learning techniques with interpretable analysis, this study provided actionable insights for optimizing SCF credit risk assessment. Firstly, a novel credit risk assessment indicator system was designed. It includes three key entities: core enterprise, SMEs, and supply chain. Secondly, based on XGBoost, CNN, RF, and LSSVM models, empirical analysis was utilized on risk assessment of 851 enterprise samples, and the experimental results of the models were compared. The primary findings are as follows:

(1) Among the evaluated models, XGBoost demonstrated the highest accuracy and stability in predicting credit risk. Its strong feature selection and anti-overfitting capabilities made it particularly well-suited for this application. This model provided robust decision-making support for financial institutions and supply chain stakeholders.

(2) To address the black-box problem of machine learning models, this study innovatively applied the SHAP algorithm to explain the XGBoost model. The analysis revealed that the asset-liability ratio positively impacts the identification of defaulting enterprises, while higher cash and quick ratios may have adverse effects. By offering both local and global interpretability, the SHAP algorithm facilitated multi-angle explanations of model predictions.

(3) The effectiveness of SHAP analysis was validated through ablation experiments, which confirmed the critical role of financial indicators such as the asset-liability ratio and quick ratio in influencing model outcomes. Based on the above conclusions, this study proposed implications of supply chain finance management.

However, this study also has some limitations. The delayed adoption of supply chain finance in China and insufficient sample enterprises may hinder model accuracy on small datasets. Future work aims to enhance accuracy by expanding sample sizes, optimizing indicators and parameters based on model interpretations. Ultimately it will focus on strengthening decision-making, promoting risk management, and propelling sustainable development of supply chain finance.

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